# Comparative Evaluation of Morphological Feature Classification Methods

**Martijn Bernard Straatsburg**
Stockholm Univeristy
mast6867@student.su.se
GitHub Repository

## Abstract

We compare the Perceptron and Multinomial Logistic Regression on morphological feature classification for the Swedish language, using data from the UniMorph database. Both models use character prefix and suffix features of lengths $1$ to $4$, combined with part-of-speech tags. After training for five epochs, Multinomial Logistic Regression achieves a token accuracy of $69.40\%$ against $65.44\%$ for the Perceptron. A paired bootstrap test with $R$=10,000 further confirms that the accuracy gap is statistically significant with $p < 0.0001$. Also, the increase in accuracy comes with a 3.4-fold increase in training time.

## 1 Introduction

Morphological feature classification is when you map a surface word form and a part-of-speech (POS) tag to a set of morphosyntactic properties, e.g. number, case, tense, and voice. In the annotation scheme of UniMorph, these different properties are encoded as semicolon separated attribute strings (Kirov et al., 2018). Good and accurate morphological analyses are required for later tasks which include parsing, machine translation, aswell as morphological reinflection.

Both the Perceptron and Multinomial Logistic Regression (MLR) are complementary approaches for multiclass linear classification. While the Perceptron uses fast and error-driven weight updates, the MLR uses and minimises a smooth cross-entropy objective and thus receives gradient information of each class on every step update. Both these models make use of the same feature representation; thus making them suitable for a comparative evaluation.

The aim of this paper is to evaluate both classifiers on the Swedish language, which is a language with rich inflectional morphology, mostly concentrated in its suffixes. We will quantify the accuracy gap, further assess its statistical significance via the paired bootstrap test, implemented from scratch, and lastly examine the highest weighted features to analyse what each model learns from surface form cues.

## 2 Data

We are using the Swedish set of the UniMorph database. Each instance is consisting of a surface word form and a morphological tag of its form `POS;FEAT`$_1$`;...;FEAT%` $_k$, where the POS tag and the feature string are both treated as separate. Multiword entries whose form contains spaces are first preprocessed by replacing them with underscores before extracting the features.

Furthermore, the training set contains $129{,}198$ instances (`swe-train`) and the test set contains $2{,}500$ instances (`swe-test`). The training set has $43$ unique morphological feature strings, which also constitute the label set for both models. Also, the test set is solely used for evaluation and plays absolutely no role in training, finetuning, or most optimal hyperparameter selection.

## 3 Methodology

### 3.1 Feature representation

For word form $w$ and candidate tag $y$, their features are created through combining a character prefix or suffix of $w$ of length $1$ to $4$ with its POS tag and $y$, thus yielding strings of the form `POS prefix=X tag=Y` or `POS suffix=X tag=Y`. So, each (word, candidate class) pair has at most eight binary features, represented as set of active feature strings; all other features are zero.

### 3.2 Perceptron

The Perceptron keeps one integer weight per (feature, class) pair which is initialised to zero. At each training instance it predicts $\hat{y} = \arg\max_y \sum_{f \in \phi(w,y)} w_{f,y}$. So, when $\hat{y} \neq y^*$, the weights for features of the gold class are plussed by 1 and weights for features of the predicted class are minused by 1. Also, the learning rate is fixed at $r = 1$.

### 3.3 Logistic Regression

The MLR is trained with stochastic gradient ascent on its log-likelihood. So, at each step, softmax probabilites are calculated over all candidate classes and each feature weight is updated via $\eta(\mathbf{1}[\text{class} = y^*] - p_y)$, where $p_y$ is the softmax probability of class $y$. Furthermore, L2 regularisation is applied only when a weight is accessed, with a strength of $\lambda = 0.0001$ and a learning rate of $\eta = 0.2$ (Jurafsky and Martin, 2026).

### 3.4 Training and evaluation

The models are trained for five epochs with not shuffling the training data. The primary evaluation metric is the proportion of correct test instances for which the predicted tag exactly matches its gold tag (token accuracy). Also, the statistical significance is measured via the one-sided paired bootstrap test with $R = 10{,}000$ resamples, thus testing whether one of the model's accuracy is better than the other's beyond chance (Efron and Tibshirani, 1995).

## 4 Results & Discussion

Table 1 show the token accuracy aswell as runtime for both models. The results show that MLR performs 3.96% better than the Perceptron. Furthermore, for the hypothesis that MLR is more accurate than the Perceptron, the paired bootstrap test results in a $p < 0.0001$; thus confirming statstical significance at $\alpha = 0.05$. The training time for MLR is 291.8 seconds compared to 86.1 seconds for the Perceptron, which comes down to a 3.4-fold increase in training time. However, evaluation times are comparable since both models use linear argmax at inference.

| Model | Acc. | Train (s) | Eval (s) |
|---|---|---|---|
| Perceptron | 65.44% | 86.1 | 0.3 |
| MLR ($\eta$=0.2, $\lambda$=$10^{-4}$) | 69.40% | 291.8 | 0.5 |
| $\Delta$ (MLR − Perceptron) | +3.96% | – | – |

Table 1: Token accuracy and runtime for both models after training for five epochs on 129,198 instances and testing on 2,500. The accuracy is statistically significant with paired bootstrap resulting in $p < 0.0001$ with $R$=10,000.

Table 2 show the five highest weighted features for each model. From which we can see some overlap, both models give high weights to adjectival suffixes *-a* (POS;PL;INDF), *-e* (POS;MASC;SG;DEF), and *-t* (POS;NEUT;SG;INDF), aswell as the nominal genitive suffix *-ets* (GEN;SG;DEF). This is similar with Swedish using suffices as its primary mean of marking inflectional distinctions. However, the main difference is that MLR ranks the nominal genitive *-s* (GEN;SG;INDF) at first, but the Perceptron ranks it nineth (see Table 3 of the Appendix).

| Perceptron | | | MLR | | |
|---|---|---|---|---|---|
| $w$ | POS | Suf / Tag | $w$ | POS | Suf / Tag |
| 21 | adj | *-a* / POS;PL;INDF | 5.97 | n | *-s* / GEN;SG;INDF |
| 20 | adj | *-e* / POS;MASC;SG;DEF | 5.11 | adj | *-e* / POS;MASC;SG;DEF |
| 19 | n | *-ets* / GEN;SG;DEF | 5.06 | adj | *-t* / POS;NEUT;SG;INDF |
| 18 | adj | *-t* / POS;NEUT;SG;INDF | 4.96 | adj | *-a* / POS;PL;INDF |
| 18 | n | *-et* / NOM;SG;DEF | 4.79 | v | *-e* / SBJV;ACT;PRS |

Table 2: Top 5 features by weight for both models. Perceptron weights are integers, while MLR weights are real-valued and L2 regularised.

The gap in accuracy most likely reflects the use of MLR's soft gradient information, namely on every trainig step MLR adjusts weights for all candidate classes proportionally to their softmax probabilities, whereas the Perceptron simply updates the gold and predicted classes when an error occurs. Furthermore, the L2 regularisation applied by the MLR limits weight growth on frequent but not perfect predicted features, of which there is no counterpart in the Perceptron.

The current results only show the token accuracy which treats all class confusions at an equal cost. However, with 43 classes, a confusion between two genitive singular forms is a smaller morphological error than a full misclassification of a verbal form as a nominal form. So, while a full breakdown of errors by POS category or confusion type isn't available, these analyses would be useful for characterising the actual qualitative differences between both models.

## 5 Conclusion & Limitations

So, the MLR results in a token accuracy of 69.40% on Swedish morphological feature classification which outperforms the Perceptron with an accuracy of 65.44% by a statistically significant difference. However, the accuracy comes at the cost of a 3.4-fold increase in training time, which might be relevant in computational performance sensitive settings.

Since both models have the same feature design and representation, the performance difference actually reflects the difference in both algorithms rather than their access to different information. However, the feature design itself sets a specific standard for both models. More so, by encoding the candidate tag directly within each feature string, its representation can't share statistical strength across morphologically similar classes. Furthermore, the set length of at most four characters for suffices may not capture the full inflectional span of any compounded or prefixed forms.

Lastly, further limitations which need to be expanded upon for future research, are to make use of any form of cross validation, instead of only having the results come from a fixed split. Furthermore, to expand on the evaluation metrics instead of solely using the token accuracy, to get a better look into qualitative differences of both models. Plus, looking into other languages than Swedish; thus further expanding on the results and conclusions of both models' generalisability on languages with more or less rich inflectional morphology.

## References


Bradley Efron and Robert Tibshirani. 1995. An introduction to the bootstrap. In *Monographs on Statistics and Applied Probability 57*.

Daniel Jurafsky and James H. Martin. 2026. *Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition with Language Models*, 3rd edition. Online manuscript released January 6, 2026.

Christo Kirov, Ryan Cotterell, John Sylak-Glassman, Géraldine Walther, Ekaterina Vylomova, Patrick Xia, Manaal Faruqui, Sabrina J. Mielke, Arya McCarthy, Sandra Kübler, David Yarowsky, Jason Eisner, and Mans Hulden. 2018. UniMorph 2.0: Universal Morphology. In *Proceedings of the Eleventh International Conference on Language Resources and Evaluation (LREC 2018)*, Miyazaki, Japan. European Language Resources Association (ELRA).


## A Top-10 Feature Weights

Table 3 extends Table 2 with the full top-10 highest-weighted features for both models.

| **Perceptron** | | | | **MLR** | | | |
|---|---|---|---|---|---|---|---|
| $w$ | POS | Suffix | Tag | $w$ | POS | Suffix | Tag |
| 21 | adj | *-a* | POS;PL;INDF | 5.97 | n | *-s* | GEN;SG;INDF |
| 20 | adj | *-e* | POS;MASC;SG;DEF | 5.11 | adj | *-e* | POS;MASC;SG;DEF |
| 19 | n | *-ets* | GEN;SG;DEF | 5.06 | adj | *-t* | POS;NEUT;SG;INDF |
| 18 | adj | *-t* | POS;NEUT;SG;INDF | 4.96 | adj | *-a* | POS;PL;INDF |
| 18 | n | *-et* | NOM;SG;DEF | 4.79 | v | *-e* | SBJV;ACT;PRS |
| 16 | n | *-ens* | GEN;PL;DEF | 4.74 | v | *-a* | IND;PL;ACT;PRS |
| 16 | v | *-es* | SBJV;PASS;PRS | 4.48 | v | *-a* | NFIN;ACT |
| 15 | n | *-nas* | GEN;PL;DEF | 4.34 | n | *-r* | NOM;PL;INDF |
| 15 | n | *-s* | GEN;SG;INDF | 4.31 | v | *-ts* | V.CVB;PASS |
| 15 | v | *-ande* | V.PTCP;PRS | 4.28 | n | *-n* | NOM;SG;DEF |

Table 3: Top 10 features by weight for both models. Perceptron weights are integers, while MLR weights are real-valued and L2 regularised.